

Medical abbreviation bake-off

CS496 AI Engineering | Project 0 | Mediterranean Institute of Technology

Measured run: 20260914T001358897252Z UTC | TEST

Scope: three local Qwen sizes, selected to avoid paid API usage. This adapts the required top-API / cheap-API / local comparison; instructor acceptance remains to be confirmed.

01 Task and dataset

Expand an abbreviation from a short UK medical-record context into one canonical term. One model call produces one answer; a deterministic function scores it without a human or LLM judge.

Example: An AMHP coordinated the statutory mental health assessment. Asked: AMHP. Gold answer: approved mental health professional.

The project contains 50 test and 10 development items with disjoint abbreviations. Labels follow the NHS glossary [1]; contexts were synthetically drafted with AI assistance and contain no patient data. The fixed convenience split puts familiar terms in dev and rarer terms in test. It is not a random sample. The review CSV records Lassaad and Rami for all 60 labels, dated 11 September 2026.

02 Reproducible setup

Models: qwen2.5:1.5b, qwen2.5:3b, qwen2.5:7b. Ollama 0.34.0; Q4_K_M quantization in the saved model metadata.

Hardware: AMD Ryzen 5 5600X 6-Core Processor; NVIDIA GeForce RTX 3060; 15.9 GB RAM; Python 3.14.3.

Shared settings: temperature 0, output limit 64 tokens, context window 8192, timeout 180 s. Identical item order, prompt and parser; sequential requests, no retries, retrieval, tools or chat history. Gold labels and vocabulary are hidden from the models.

Prompt instruction: Expand the medical abbreviation in the supplied UK medical-record context. Return ONLY the full medical term on one line. No abbreviation, explanation, quotes, JSON, Answer: label or final full stop. Use the standard UK expansion indicated by context. Full exact prompts are saved with the run.

03 Results from all 150 attempts

Model	Correct / accuracy	p50 / p95 ms	USD / 1k	Tokens/s	P / R / T / O
qwen2.5:1.5b	16/50 (32%)	2228 / 2413	5.0950	50.6	33 / 0 / 0 / 0
qwen2.5:3b	16/50 (32%)	2223 / 2361	5.0956	58.7	33 / 0 / 0 / 0
qwen2.5:7b	27/50 (54%)	2272 / 2387	5.0989	37.0	22 / 0 / 0 / 0

P=parse, R=refusal, T=timeout, O=other execution errors. Wrong in-vocabulary answers remain in the accuracy denominator; P includes out-of-vocabulary wording, not just broken formatting. Explicit refusal prefixes are detected; other refusal wording can remain P. All failures count as wrong.

Latency is full request wall time, including any loading; pacing pauses are excluded. p50/p95 use linear interpolation over all attempts.

Tokens/s = total generation tokens / total generation seconds from Ollama [2], not end-to-end throughput.

04 Normalization and false-negative audit

The scorer trims outer whitespace and case-folds both answer and label, then requires exact wording and one line. On these same responses, case-sensitive matching would accept 3/0/0; the frozen normalized scorer accepts 16/16/27 (small/medium/large). This is a diagnostic comparison, not a new model run.

All 91 failed responses were checked in a mechanical spelling audit. 14 are spelling/spacing/hyphen candidates. They remain wrong under the frozen rule; the audit does not certify clinical equivalence or change accuracy.

Errors, decision and cost

05 Three recorded wrong answers per model

Model / item	Expected	Actual response
qwen2.5:1.5b / AMHP	approved mental health professional	Assessment and Management of People with Mental Health Problems
qwen2.5:1.5b / BNO	bowels not open	Bowel Number
qwen2.5:1.5b / BO	bowels open	bowel
qwen2.5:3b / AMHP	approved mental health professional	Adult Mental Health Practitioner
qwen2.5:3b / BNO	bowels not open	Bowel Not Operating
qwen2.5:3b / BO	bowels open	Bowel obstruction
qwen2.5:7b / BNO	bowels not open	Bowel Not Open
qwen2.5:7b / BO	bowels open	Bowel Opening
qwen2.5:7b / CSU	catheter stream urine sample	Catheterised Suprapubic Urine

First three failures per model in recorded order; all nine are parse_error under the vocabulary policy. Full outputs and all remaining errors are in results/per_item.csv.

06 What the errors mean

Capitalization alone is now accepted: Activated Partial Thromboplastin Time matches its lower-case label. Fecal versus faecal and High Density versus high-density are candidates for a future spelling policy. Investigation versus investigations is a singular/plural difference. Physiotherapy versus physiotherapist changes an activity into a professional role; blindly stemming words would conceal such distinctions.

Early dev prompts exposed the vocabulary and produced 10/10 scores. The final task hides it, so those pilots are not directly comparable. A case-sensitive 150-call run on 13 September accepted only 3 answers; a later full rerun used normalization. The archived 14 September run reproduced all 150 previous raw answer strings, but latency changed. These repeated items do not constitute 300 independent test examples.

07 Model choice and switching conditions

No model meets the project's 95% accuracy + p95<10s target. For this benchmark, choose the highest-accuracy model: qwen2.5:7b at 27/50 (54.0%).

Use the highest-accuracy model only as the benchmark preference, not as a clinical deployment recommendation. The 95% accuracy and p95 below 10 s thresholds are project decision rules, not course requirements; preregistration is not established. Switch to a smaller model if it meets those targets at lower cost on a new frozen set. Re-evaluate if hardware, traffic or wording policy changes.

08 Cost today, at 100x, and break-even

No API fees. Assumptions: \$0.15/active hardware hour, 1 labour hour/month at \$5/hour, 160 available hours/month. Baseline 1,000 requests; 100x 100,000. These are estimates, not measured invoices.

Model	Baseline USD/mo	100x USD/mo	Capacity requests/mo
qwen2.5:1.5b	5.095	14.502	252577
qwen2.5:3b	5.096	14.562	250990
qwen2.5:7b	5.099	14.893	242601

For mean request seconds s: throughput = 3600/s; variable cost v = hardware hourly rate / throughput; monthly cost = labour F + volume N*v. Each candidate is an alternative deployment. For this measured run, the capacity table supports 100x under these assumptions; pauses, always-on rental and scaling overhead are excluded. Throughput and labour are assumed unchanged.

API/local break-even would be $N = F/(A-v)$, where A is API cost/request, only when $A > v$ and capacity permits. With no measured API comparator, a numerical break-even cannot be claimed. Instructor approval is needed for this omission.

Limits: one model family, one machine, 50 synthetic test items (one item = 2 percentage points), and exact canonical wording. No statistical significance or clinical competence is established.

[1] NHS: nhs.uk/nhs-app/help/understanding-abbreviations/ (checked 14 Sep 2026). [2] Ollama: docs.ollama.com/api/chat. Run metadata, raw responses and hashes: results/runs/.